

General retinal layer segmentation in OCT images via reinforcement constraint

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ABSTRACT

The change of layer thickness of retina is closely associated with the development of ocular diseases such as glaucoma and optic disc drusen. Optical coherence tomography (OCT) is a widely used technology to visualize the lamellar structures of retina. Accurate segmentation of retinal lamellar structures is crucial for diagnosis, treatment, and related research of ocular diseases. However, existing studies have focused on improving the segmentation accuracy, they cannot achieve consistent segmentation performance on different types of datasets, such as retinal OCT images with optic disc and interference of diseases. To this end, a general retinal layer segmentation method is presented in this paper. To obtain more continuous and smoother boundaries, feature enhanced decoding module with reinforcement constraint is proposed, fusing boundary prior and distribution prior, and correcting bias in learning process simultaneously. To enhance the model's perception of the slender retinal structure, position channel attention is introduced, obtaining global dependencies of both space and channel. To handle the imbalanced distribution of retinal OCT images, focal loss is introduced, guiding the model to pay more attention to retinal layers with a smaller proportion. The designed method achieves the state-of-the-art (SOTA) overall performance on five datasets (i.e., MGU, DUKE, NR206, OCTA500 and private dataset).

1. Introduction

With the ageing of population and changing in people's lifestyles, an increasing number of people are troubled with ocular diseases. Early screening and accurate diagnosis of ocular diseases are crucial to achieve early prevention and targeted treatment (Yi et al., 2023). Many studies have demonstrated that some ocular diseases are related to the thickness of retinal layers. For instance, patients with multiple sclerosis or neuromyelitis optica may appear thinning of the ganglion cell layer (GCL) (Syc et al., 2012). Optic disc drusen can cause defects in retinal nerve fiber layer (RNFL) with progressive visual field loss (Engelke et al., 2020). The RNFL and GCL in patients with glaucoma exhibit a progressive reduction in thickness over time (Hood et al., 2013). In this case, the structural change in retinal layer has become an important basis for ophthalmologists to diagnose ocular diseases.

The retina is a lamellar structure situated within the posterior wall of the eye. In clinical applications, the optical coherence tomography (OCT) is served as a non-invasive tool to display various layers of retina (Nawaz et al., 2023), as shown in Fig. 1(a). Fig. 1(b) shows the labels of each layer annotated by professional ophthalmologists. Except

for the centered optic disc, the retina has a complex anatomical structure with nine layers, named retinal nerve fiber layer (RNFL), ganglion cell layer (GCL), inner plexiform layer (IPL), inner nuclear layer (INL), outer plexiform layer (OPL), outer nuclear layer (ONL), inner/outer photoreceptor segment (IS/OS), retinal pigment epithelium (RPE), and choroid (Li et al., 2021). Due to the interference of noise, lesions, and blurred boundaries among retinal layers, it is very time-consuming for ophthalmologists to distinguish each layer without computer aided analysis, resulting in difficulties for analyzing the correlations between retinal layers and ocular diseases. Therefore, it is of great significance to segment retinal layers accurately for intelligent computer-aided diagnosis of ocular diseases.

In recent years, a few methods have been proposed for retinal layer segmentation. Duan et al. (2018) proposed a generative segmentation method based on groupwise curve alignment to segment different retinal layers. Lou et al. (2023) utilized an improved graph theory algorithm to correct the segmentation boundaries extracted by bidirectional wave algorithm, achieving the segmentation of 8-layer retinal boundaries. Yu et al. (2018) proposed an improved graph search algorithm to handle the segmentation of retinal OCT images with optic

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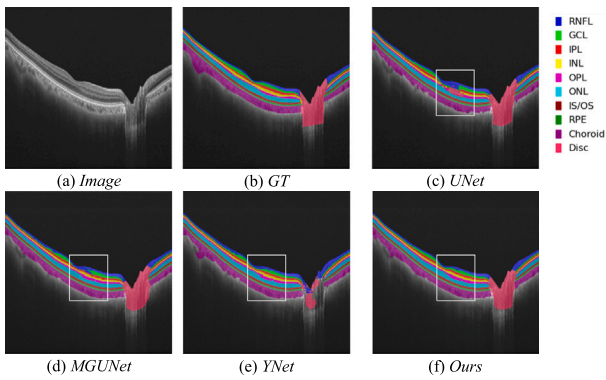


Fig. 1. An example of OCT image. (a) Retinal lamellar structures. (b) The labels of each layer. (c) UNet's prediction. (d) MGUNet's prediction. (e) YNet's prediction. (f) Ours.

disc. Novosel et al. (2015) and Liu et al. (2018) utilized level set methods to achieve multilayer segmentation while ensuring sequential layer structures. Rossant et al. (2015) proposed Parallel Double Snakes, which allows two curves to evolve simultaneously under parallel constraint, and it achieves better adaptation to the slender structure of retinal layers. These methods use traditional image processing methods that fall short of high segmentation accuracy and robust model generalization.

At present, many researchers have also applied deep learning methods to retinal layer segmentation. Roy et al. (2017) used a fully convolutional network to segment retinal layers and fluid. He et al. (2023) adopted ConvNeXt as the backbone and improved it with a multi-scale feature extraction module and a depth-efficient convolutional attention module to capture a broader spectrum of features. Farshad et al. (2022) introduced a spatial-spectral encoder to provide larger receptive fields. As convolutional neural network (CNN) is limited to extracting global features (Li et al., 2020), the transformer-based methods exhibit a better ability to extract global features (Lu et al., 2024b; Huang et al., 2024b,a; Lu et al., 2024a; Deng et al., 2025; Sun et al., 2024; Chen et al., 2024). Xue et al. (2023) and Zhang et al. (2023) combined CNN and Vision Transformer (ViT) to improve the global modeling ability of their methods. These methods acknowledge that the CNN lacks the ability to extract global information, thus aim to enhance the perception of the retina. However, the problem of blurred boundaries still remains. In this case, Wang et al. (2021) proposed a boundary-aware U-Net, improving the segmentation performance by detecting accurate boundaries. Meanwhile, a topology guarantee loss was used to ensure the sequential structure of retinal layers. He et al. (2021) modeled the distribution of surface positions to obtain continuous and correct boundaries. The above two methods predict the boundaries in a top-down manner iteratively, which suffer from a risk of affecting latter boundary prediction with the wrong former boundary prediction. In this case, Tan et al. (2023) and Shen et al. (2023) combined topological constraint and mean squared error loss, prompting the model to obtain more continuous and smoother results, but it did not work in OCT images with optic disc. Thus, Li et al. (2021) designed a two-stage network specialized for segmenting optic disc. However, the distribution imbalance among retinal layers was not considered. To handle this problem, Yang and Farsiu (2023) used connectivity labels to obtain directional information. Then, a weighted loss based on label distribution was proposed to give more attention to retinal layers that are difficult to segment.

Although previous methods have achieved acceptable segmentation performance, they do not exhibit good robustness simultaneously on various types of datasets, such as retinal OCT images with optic disc and lesion interference. In addition, many research prediction results do not achieve good continuity and smoothness. As shown in the white box of

Fig. 1(c) (e), the prediction results of UNet and MGUNet are broken in the GCL layer. In addition, the prediction results of many researches do not achieve good continuity and smoothness. As shown in the white boxes of Fig. 1(c) (d), the prediction results of UNet (Ronneberger et al., 2015) and MGUNet (Li et al., 2021) are broken in the GCL layer, and UNet leads to a large outlier that is misidentified as optic disc. Therefore, the continuity of the prediction results of these two methods is not good. In the white box of Fig. 1(e), although YNet (Farshad et al., 2022) gets better continuity, the prediction results have more spikes, resulting in unsatisfactory smoothness. Therefore, we design a general model for retinal layer segmentation in this article. To obtain more continuous and smoother boundaries, a feature enhanced decoding module with reinforcement constraint (RC-FEDM) is proposed. The RC-FEDM fuses boundary prior and distribution prior so that the model pays more attention to the retinal boundary information and obtains the distribution information of different regions. Moreover, reinforcement constraint is used to correct bias in the learning process. To enhance the model's perception of the slender structure of retina, position channel attention is introduced, which not only establishes the dependencies of distant pixels but also models the dependencies between channels. By this means, the global information of both space and channel is obtained. To handle the imbalanced distribution of retinal OCT images, focal loss is introduced, guiding the model to pay more attention to retinal layers with a smaller proportion. The proposed method achieves the state-of-the-art overall performance on five datasets. As shown in Fig. 1(f), our method has a good continuity and smoothness.

2. Method

2.1. Framework overview

In clinical practice, retinal OCT images with the optic disc are usually employed to diagnose ocular diseases. Compared with retina, optic disc has a different structure, which interferes with the segmentation performance of the model. Inspired by Li et al. (2021), we employ a two-stage segmentation paradigm, reducing the interference from optic disc. As shown in Fig. 2(a), the overall framework is composed of the following steps: optic disc segmentation, retina segmentation, and feature fusion. The training process is as follows. Firstly, given the input OCT images, the optic disc is partitioned from the retina. Subsequently, the retinal OCT images without optic disc are implemented to segment the retinal layers. Lastly, the features of optic disc and retinal layers are fused to obtain the final segmentation result. Specifically, the feature fusion module concatenates the features of optic disc and retinal layers together in channel dimension, then the features pass through a Softmax to obtain the final segmentation result. Concurrently, the distribution comparison modules of both optic disc and retinal segmentation networks, along with the focal loss and Dice loss, are used to guide the model's learning.

2.2. Segmentation network

Due to the difficulty in obtaining numerous labeled data for OCT images, the retinal layer segmentation usually lacks samples for training (Pestiev et al., 2019). Since the U-Net architecture (Ronneberger et al., 2015) achieves successful performance on small-scale datasets and is widely employed in various medical image segmentation tasks (Parra-Mora and da Silva Cruz, 2022), the encoder-decoder architecture is used in our study. As shown in Fig. 2(b), the segmentation network comprises three components: encoder, attention module, and decoder with reinforcement constraint. Skip connections are adopted to connect the encoder and decoder, facilitating the integration of low-level and high-level features.

2.2.1. Encoder

As shown in Fig. 2(b), the encoder comprises four encoding modules, which are displayed in detail in Fig. 2(c) and (d). The structure of

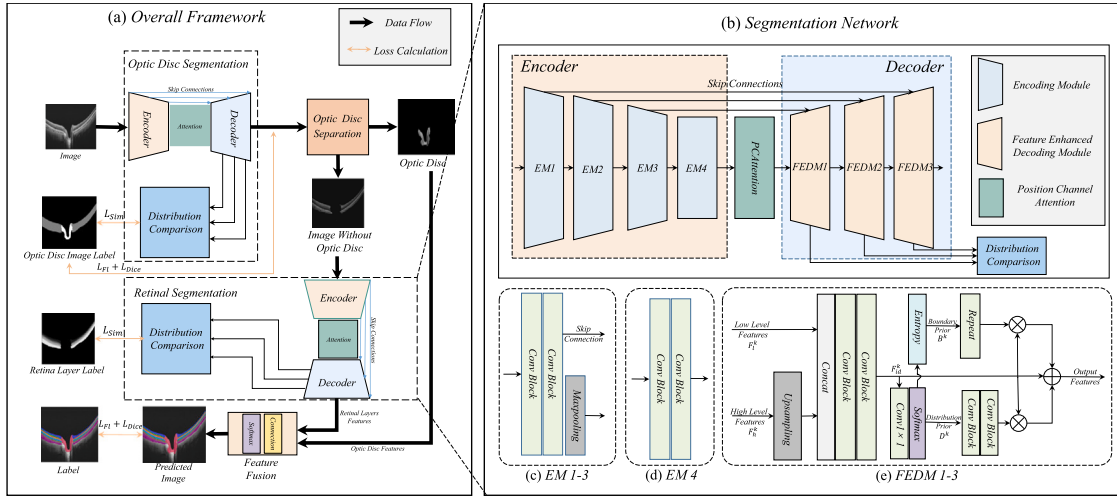


Fig. 2. The framework of our method. (a) The overall framework. (b) The structure of segmentation network. (c) Detailed structure of encoding module 1–3. (d) Detailed structure of encoding module 4. (e) Detailed structure of feature enhanced decoding module.

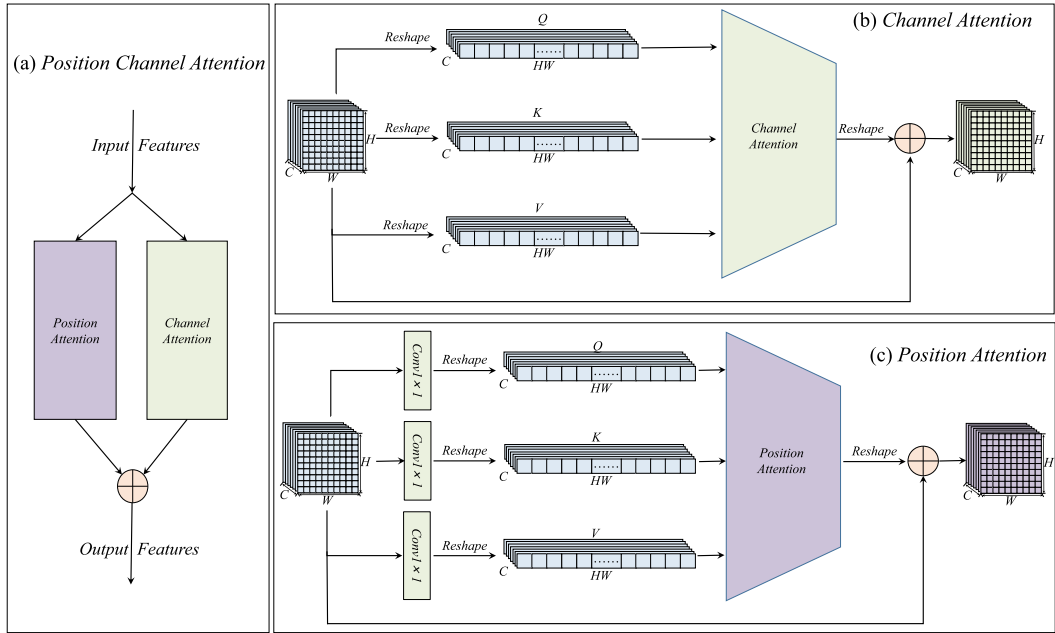


Fig. 3. The structure of attention module. (a) The position channel attention. (b) Detailed structure of position attention. (c) Detailed structure of channel attention.

Conv Block in each encoding module is expressed as follows:

$$x^l = \text{ReLU}(\text{BN}(\text{Conv}(x^{l-1}))) \quad (1)$$

where x^{l-1} is the output of the $l-1$ th layer, x^l is the output of the l th layer; and Conv represents convolution operation, whose kernel size is 3. BN denotes batch normalization.

For optic disc segmentation, we configure the base channel (BC) to 16. The output channels of Conv Block are set to BC , $2 \times BC$, $4 \times BC$, and $8 \times BC$, with downsampling kernel sizes of 2, 4, and 4, respectively. For retinal segmentation, due to the small thickness of each retinal layer, large downsampling kernels may lead to the loss of detailed information. In this case, all downsampling kernel sizes are adjusted to 2.

2.2.2. Attention module

Although the CNN employed in the encoder can capture local information for retinal layer segmentation, due to the slender structures and expansive span of each retinal layer, global information is required

to augment the perceptive ability of the model. Inspired by [Fu et al. \(2019\)](#), the position channel attention module is involved to model global dependencies of both space and channel.

The position channel attention comprises two parts: position attention and channel attention. Position attention obtains dependencies between pixels, getting global information spatially. As shown in [Fig. 3\(b\)](#), firstly, the input feature maps, whose size is $(C \times H \times W)$, pass through a convolution layer with a kernel size of 1×1 , followed by reshaping the resulting matrix to dimensions $(C \times HW)$ to obtain queries (Q), keys (K), and values (V). Then, the output matrix is computed by Eq. (2), subsequently reshaped to $(C \times H \times W)$. Finally, the reshaped matrix is added to the original features, getting the position attention features.

$$P\text{Attention}(Q, K, V) = V \text{Softmax}(Q^T K) \quad (2)$$

Channel attention models dependencies between channels, facilitating the extraction of global information about the channels. As shown in [Fig. 3\(c\)](#), firstly, the input feature maps, whose size is $(C \times H \times W)$,

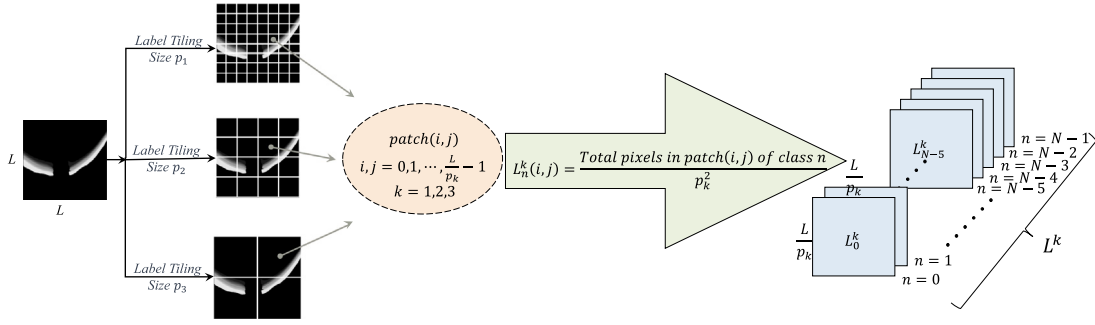


Fig. 4. The acquisition of regional label distribution.

are reshaped to dimensions $(C \times HW)$ to obtain queries (Q), keys (K), and values (V). Then, the output matrix is computed by Eq. (3). Finally, the channel attention features are obtained by reshaping the matrix to $(C \times H \times W)$ and adding it to the original features.

$$CAttention(Q, K, V) = Softmax(QK^T)V \quad (3)$$

As shown in Fig. 3(a), the final features are obtained by combining the acquired position attention features and channel attention features.

2.2.3. Decoder with reinforcement constraint

Down-sampling leads to the loss of detailed information, and up-sampling could introduce boundary blurring, consequently affecting the precision of retinal layer segmentation. While skip connections integrate low-level and high-level features to handle the above issues, they are not enough for the following two reasons: (1) In contrast to natural images, the adjacent retinal layers exhibit high similarity, with only subtle differences, which brings much difficulty for determining the final segmentation result. (2) The presence of distinct boundaries in retinal OCT images, especially in areas affected by lesion interference, also brings challenges for boundary segmentation of retinal layers. Inspired by Li et al. (2023), we handle these problems by incorporating distribution and boundary prior to the decoding modules.

As shown in Fig. 2(b), three successive feature enhanced decoding modules (FEDMs) are used, restoring the extracted spatial features step by step. The structure of FEDM is shown in Fig. 2(e), the input features contain low-level features from the skip connections and high-level features obtained from the position channel attention, denoted by F_l^k and F_h^k , respectively. The low-level features are concatenated with up-sampled high-level features, and then processed by two Conv Blocks, generating the initial decoding features, denoted as F_{id}^k . Subsequently, the distribution prior is obtained from the initial decoding features F_{id}^k , expressed as follows:

$$D^k = Softmax(Conv(F_{id}^k)) \quad (4)$$

where $D^k \in R^{B \times N \times H_k \times W_k}$. N represents the number of classes, and B denotes the batch size. H_k and W_k represent the height and width of initial decoding features F_{id}^k respectively.

Identifying pixels at the ambiguous boundary of retinal layers is very challenging, since they carry more information with high uncertainty compared with other non-boundary pixels. To effectively identify the boundary pixels, we compute the information entropy of distribution characteristics, extracting the boundary prior as follows:

$$B^k(b, h, w) = - \sum_{n=0}^{N-1} D^k(b, n, h, w) \log D^k(b, n, h, w) \quad (5)$$

where $B^k \in R^{B \times H_k \times W_k}$.

The final output features are acquired by combining the boundary prior and distribution prior with the initial decoding features through fusion process. Firstly, the boundary prior is repeated along the channel dimension and multiplied with initial decoding features F_{id}^k as expressed in Eq. (6), obtaining boundary-enhanced features F_B^k , where

$*$ denotes the element-wise multiplication operation. Secondly, the distribution prior is processed by two Conv Blocks and multiplied with F_{id}^k as expressed in Eq. (7), obtaining distribution-enhanced features F_D^k . Finally, the output features F_O^k are acquired by the sum of F_B^k , F_D^k and F_{id}^k , as expressed in Eq. (8).

$$F_B^k = Repeats(B^k) * F_{id}^k \quad (6)$$

$$F_D^k = ConvBlock(ConvBlock(D^k)) * F_{id}^k \quad (7)$$

$$F_O^k = F_B^k + F_D^k + F_{id}^k \quad (8)$$

2.2.4. Distribution comparison module

Due to the inherent structural characteristics of fundus OCT images, as illustrated in Fig. 1. Different layers exhibit a strict sequential organization. For instance, the RNFL is always positioned at the top layer, the Choroid at the bottom, and the retina layers surround the optic disc. By incorporating label distribution information into the model, we enable it to more effectively learn and utilize this prior knowledge, enhancing its understanding of the underlying anatomical structure.

The fusion of prior information effectively complements the detailed information. However, to achieve continuous and smooth boundaries, imposing constraint on prior information is necessary. Reinforcement constraint is used to obtain more accurate prior information and reduce the prior bias in the learning process. Specifically, the distribution prior D^k obtained by Eq. (4) is compared with the corresponding regional label distribution by the distribution comparison module. Specifically, as shown in Fig. 2(e). The output of softmax and the label distribution obtained by label statistics are used to calculate the KL divergence to constrain the distribution prior.

The acquisition of regional label distribution is shown in Fig. 4. Firstly, the input labels are cut at various scales, with patch lengths of p_k , where $k = 1, 2, 3$. For optic disc segmentation, p_k is adjusted to 1, 2, 8, respectively. For segmentation of retinal layers, p_k is set to 1, 2, 4, respectively. For the k th group of label patches, the patch with coordinate (i, j) is represented as $patch(i, j)$, where $i, j = 0, 1, \dots, \frac{L}{p_k} - 1$, and L denotes the image size. By calculating the probability distribution in each patch using Eq. (9), the probability distribution map of the n th class can be obtained, denoted as L_n^k , whose size is $\frac{L}{p_k} \times \frac{L}{p_k}$.

$$L_n^k(i, j) = \frac{\text{Total pixels in patch}(i, j) \text{ of class } n}{p_k^2} \quad (9)$$

After obtaining the regional label distribution L^k , which is the combination of L_n^k of all classes, we compute the similarity between L^k and the distribution prior D^k using Kullback–Leibler divergence as follows:

$$L_{Sim} = \frac{1}{3} \sum_{k=1}^3 D_{KL}(D^k \parallel L^k) \quad (10)$$

Table 1
Segmentation performance on MGU dataset by different methods (%).

	Dice/(PA)							
	U-Net (Ronneberger et al., 2015)	ReLayNet (Roy et al., 2017)	MGUNet (Li et al., 2021)	Y-Net (Farshad et al., 2022)	EMVNet (He et al., 2023)	DA-TransUNet (Sun et al., 2024)	TransUNet (Chen et al., 2024)	Ours
RNFL	81.97 (83.19)	82.22 (84.23)	82.40 (85.31)	81.67 (83.67)	81.40 (82.32)	82.34 (83.19)	82.75 (84.83)	83.52 (84.59)
GCL	67.67 (67.93)	67.64 (64.04)	68.91 (68.00)	67.17 (66.59)	65.96 (62.95)	68.05 (67.52)	66.74 (67.90)	71.22 (72.34)
IPL	72.32 (71.70)	74.31 (77.37)	74.07 (72.44)	71.80 (69.49)	71.06 (75.47)	74.05 (75.04)	71.82 (70.38)	74.99 (73.53)
INL	76.84 (78.03)	76.71 (75.14)	79.13 (80.08)	76.38 (76.21)	77.73 (79.79)	77.34 (77.65)	76.15 (76.04)	79.55 (80.35)
OPL	80.36 (79.59)	80.69 (80.29)	82.23 (83.03)	80.80 (79.38)	77.96 (80.54)	80.84 (80.83)	80.39 (79.61)	82.29 (82.01)
ONL	90.64 (91.33)	90.44 (90.81)	91.07 (91.23)	90.69 (92.00)	90.41 (90.51)	90.66 (90.43)	90.48 (90.14)	91.23 (93.20)
IS/OS	86.00 (84.56)	86.37 (85.96)	86.65 (87.47)	86.49 (85.96)	86.05 (87.07)	86.34 (87.36)	86.21 (87.75)	86.57 (86.48)
RPE	82.66 (83.85)	88.22 (81.64)	82.92 (83.01)	82.66 (82.78)	82.02 (83.19)	82.56 (84.76)	82.53 (84.70)	82.95 (84.01)
Choroid	88.56 (87.07)	88.22 (90.01)	89.04 (88.46)	88.07 (88.56)	87.80 (89.66)	88.33 (85.74)	88.95 (88.91)	89.60 (90.06)
Disc	79.62 (80.26)	77.08 (81.96)	82.54 (88.48)	79.11 (83.28)	80.89 (89.16)	82.57 (87.46)	81.01 (87.02)	84.49 (88.44)
Layer	80.78 (80.81)	81.65 (81.05)	81.82 (82.11)	80.64 (80.52)	80.04 (81.28)	81.17 (81.39)	80.67 (81.14)	82.44 (82.95)
Average	80.66 (80.75)	80.59 (81.14)	81.90 (82.75)	80.48 (80.79)	80.13 (82.07)	81.31 (82.00)	81.01 (81.73)	82.64 (83.50)

2.2.5. Loss function

Considering the distribution imbalance among retinal layers in OCT images, inspired by Lin et al. (2017), focal loss is employed to handle this issue, guiding the model's attention to retinal layers that are difficult to segment. The focal loss is expressed as:

$$L_{Fl} = -\frac{1}{N} \sum_{n=0}^{N-1} \alpha(1-p_n)^\gamma \log(p_n) + (1-\alpha)(p_n)^\gamma \log(1-p_n) \quad (11)$$

where p_n represents the probability assigned by the model to predict a positive sample in the class n . α and γ are adjustment factors. We set $\alpha = 0.25$ and $\gamma = 2$ empirically.

To align the segmentation results with actual labels, Dice loss (Lu et al., 2023b) is used, which is expressed as follows:

$$L_{Dice} = 1 - \frac{2}{N-1} \sum_{n=1}^{N-1} \frac{|X^n \cap Y^n|}{|X^n| + |Y^n|} \quad (12)$$

where X^n represents the segmented area in class n , and Y^n represents the ground truth in class n .

It can be seen from Fig. 2(a) that the loss used for segmentation of optic disc (L_{OD}) and the loss used for segmentation of retinal layers (L_{RE}) are both sum of L_{Fl} , L_{Dice} , and L_{Sim} . Following MGUNet (Li et al., 2021), the total loss is the weighted sum of L_{OD} and L_{RE} expressed as follows:

$$L_{total} = L_{OD} + 2L_{RE} \quad (13)$$

3. Experiments and results

3.1. Datasets and comparison models

To assess the effectiveness of our designed model, experiments were conducted on five datasets: MGU dataset (Li et al., 2021), DUKE dataset (Chiu et al., 2015), NR206 dataset (He et al., 2023), OCTA500 dataset (Li et al., 2024) and private dataset. MGU dataset includes images from 61 subjects, each containing 12 B-scans. The image size

is 1024×992 . The labels contain optic disc and 9-layer annotations of the retina. DUKE dataset comprises 110 B-scans from 10 patients with diabetic macular edema. The images are of size 496×768 , and the labels include annotations for 7-layer structures of the retina. NR206 dataset contains 206 B-scans of healthy human, with the image size of 500×750 , and the labels include annotations for 8-layer structures of the retina. The OCTA500 dataset consists of 3D data from 300 patients. For each patient, the data is sampled at intervals to generate 20 two-dimensional OCT slices in size of, each measuring 600×400 pixels. This resulted in containing a total of 6,000 images. Private dataset comprises 100 B-scans from 100 patients. The image size is 512×920 . The labels contain optic disc, retina and choroid. Comparison study was performed with U-Net (Ronneberger et al., 2015), ReLayNet (Roy et al., 2017), MGUNet (Li et al., 2021), Y-Net (Farshad et al., 2022), EMVNet (He et al., 2023), DA-TransUNet (Sun et al., 2024) and TransUNet (Chen et al., 2024) to demonstrate the superiority of our method.

3.2. Implementation detail and evaluation metrics

In our work, all methods were implemented in PyTorch and trained on NVIDIA RTX 3090 GPU. The Adam (Kingma and Ba, 2014) optimizer with default parameters was utilized to update the model weights, employing a batch size of 2. All models were trained for 100 epochs, and the model with the best performance on validation set was selected for testing. The initial learning rate was set to 0.001 and scaled down tenfold every 20 epochs. During the evaluation phase, Dice coefficient (Dice) (Lu et al., 2023a) and pixel accuracy (PA) (Li et al., 2021) were utilized to measure the segmentation performance quantitatively.

3.3. Main results

3.3.1. Quantitative analysis

The performance on three datasets is analyzed separately. MGU dataset includes both the optic disc and the retina. Therefore, we

Table 2
Segmentation performance on DUKE dataset by different methods (%).

	Dice/(PA)							
	U-Net (Ronneberger et al., 2015)	ReLayNet (Roy et al., 2017)	MGUNet (Li et al., 2021)	Y-Net (Farshad et al., 2022)	EMVNet (He et al., 2023)	DA-TransUNet (Sun et al., 2024)	TransUNet (Chen et al., 2024)	Ours
RNFL	87.09 (87.81)	87.71 (89.34)	86.67 (86.51)	86.47 (87.14)	83.29 (86.30)	85.42 (84.87)	86.19 (86.74)	87.80 (89.23)
GCL+IPL	87.07 (89.67)	90.29 (90.68)	90.11 (91.86)	88.60 (89.85)	88.04 (89.31)	89.11 (91.32)	89.80 (90.94)	91.07 (91.48)
INL	69.93 (81.18)	80.38 (82.32)	79.96 (82.90)	78.04 (85.45)	74.48 (80.49)	80.11 (83.64)	80.92 (83.60)	81.40 (84.14)
OPL	75.43 (76.41)	79.04 (78.74)	79.43 (79.79)	69.79 (60.11)	77.09 (77.64)	79.82 (79.54)	79.83 (80.17)	80.36 (81.67)
ONL+ISM	89.97 (91.71)	93.27 (92.57)	94.03 (93.44)	92.27 (94.33)	90.71 (91.02)	93.59 (92.97)	94.02 (93.51)	93.77 (92.47)
ISE	89.74 (90.52)	89.26 (89.95)	89.40 (90.59)	88.85 (89.65)	89.12 (90.30)	89.19 (89.53)	88.95 (90.67)	89.43 (89.37)
OS-RPE	88.11 (87.54)	87.84 (87.80)	87.90 (87.29)	87.83 (87.96)	87.44 (90.47)	87.74 (88.42)	87.84 (88.11)	88.13 (88.72)
Average	83.90 (86.41)	86.83 (87.34)	86.79 (87.48)	84.55 (84.93)	84.31 (86.50)	86.43 (87.19)	86.79 (87.68)	87.42 (88.15)

Table 3
Segmentation performance on NR206 dataset by different methods (%).

	Dice/(PA)							
	U-Net (Ronneberger et al., 2015)	ReLayNet (Roy et al., 2017)	MGUNet (Li et al., 2021)	Y-Net (Farshad et al., 2022)	EMVNet (He et al., 2023)	DA-TransUNet (Sun et al., 2024)	TransUNet (Chen et al., 2024)	Ours
RNFL	90.28 (90.91)	88.98 (89.74)	90.55 (91.08)	90.00 (91.28)	88.46 (92.76)	90.79 (92.98)	90.41 (94.01)	90.97 (91.46)
GCL+IPL	96.04 (95.52)	94.97 (94.83)	96.37 (96.25)	96.42 (96.40)	96.32 (95.71)	96.31 (95.40)	96.17 (95.43)	96.53 (96.22)
INL	90.48 (90.95)	88.49 (88.73)	90.99 (91.26)	91.11 (91.82)	90.63 (92.28)	90.76 (91.67)	90.70 (91.13)	91.39 (92.25)
OPL	82.25 (84.17)	79.22 (80.06)	82.84 (83.18)	82.55 (82.24)	82.97 (81.33)	82.35 (82.55)	82.40 (82.27)	83.38 (83.94)
ONL	95.23 (95.40)	94.61 (94.80)	95.33 (95.34)	95.37 (95.61)	95.72 (96.96)	95.42 (96.19)	95.39 (95.98)	95.58 (95.84)
IS/OS	92.78 (92.13)	92.46 (92.11)	92.44 (91.68)	92.42 (92.24)	92.66 (92.43)	92.69 (91.96)	92.66 (91.93)	92.97 (92.62)
RPE	88.24 (88.78)	87.58 (88.04)	88.05 (88.45)	87.58 (88.42)	88.31 (89.23)	87.84 (90.31)	87.56 (87.60)	88.32 (89.02)
Choroid	96.28 (95.52)	96.07 (96.60)	96.41 (96.39)	96.25 (96.48)	95.83 (97.17)	96.30 (95.58)	96.13 (95.61)	96.50 (96.59)
Average	91.45 (91.67)	90.30 (90.61)	91.62 (91.70)	91.46 (91.81)	91.36 (92.23)	91.55 (92.08)	91.43 (91.75)	91.95 (92.24)

evaluate the Dice and PA for each retinal layer, the average of all layers (Layer), and the overall average including the optic disc (Average). As shown in Table 1, the Dice and PA for Average reach 82.64% and 83.50%, surpassing the sub-optimal MGUNet by 0.74% and 0.75%. The Layer also attains the best in both metrics. For optic disc segmentation, our method achieves the optimal Dice and the third best PA. In addition, our method achieves the best Dice and PA across most layers. GCL, accounting for the minimum proportion, is the most challenging layer for segmentation. Notably, due to the focus on the perception of slender structures and the supplementation of boundary and distribution prior information, our method is significantly improved in GCL, increasing 2.31% and 4.34% in Dice and PA compared with MGUNet, respectively.

For DUKE dataset, there is no optic disc in OCT images, but our model may be affected by the lesions of fluid and edema during the segmentation of retinal layers. Fluid and edema destroy the structure of the retinal layers, making the segmentation more challenging. In this case, the base channel (*BC*) of the encoder is adjusted to 32 to give the

encoder stronger feature extraction capabilities. As shown in Table 2, our method achieves the best overall performance with 87.42% in Dice and 88.15% in PA, surpassing the sub-optimal ReLayNet by 0.59% and MGUNet by 0.67%, respectively. It is worth noting that although some methods achieve the highest PA on a certain layer, they are usually at the expense of greatly sacrificing the segmentation performance of neighboring layers. For example, Y-Net achieves 85.45% at the INL layer, but only 60.11% at the OPL layer. This shows that Y-Net misrecognizes part of the OPL as the INL. Since the proposed RC-FEDM allows the model to pay more attention to the boundaries of each layer and reduces the interference of neighboring fuzzy pixels, our approach achieves a better segmentation performance of adjacent layers, which in turn leads to the highest overall PA.

For NR206 dataset, there is also no optic disc in OCT images. The segmentation results are shown in Table 3. It can be seen that our method achieves the average Dice score of 91.95% and PA of 92.24%, which are 0.33% and 0.01% higher than the suboptimal methods,

Table 4
Segmentation performance on OCTA500 dataset by different methods (%).

	Dice/(PA)							
	U-Net (Ronneberger et al., 2015)	ReLayNet (Roy et al., 2017)	MGUNet (Li et al., 2021)	Y-Net (Farshad et al., 2022)	EMVNet (He et al., 2023)	DA-TransUNet (Sun et al., 2024)	TransUNet (Chen et al., 2024)	Ours
IML-ILP	97.68 (97.74)	97.59 (97.82)	97.67 (97.93)	96.30 (97.06)	97.44 (97.68)	97.49 (97.94)	97.74 (97.88)	97.78 (97.78)
IPL-OPL	93.91 (94.09)	93.16 (93.23)	93.57 (93.23)	89.88 (88.59)	92.57 (92.65)	94.00 (94.32)	94.15 (94.57)	94.03 (94.17)
OPL-IS/OS	95.30 (95.50)	94.50 (94.25)	94.63 (94.16)	91.42 (92.28)	94.28 (94.03)	95.20 (94.95)	95.38 (94.99)	95.39 (95.40)
IS/OS-RPE	94.01 (93.93)	93.14 (93.65)	93.44 (93.98)	85.46 (88.01)	93.58 (93.49)	93.58 (94.31)	93.82 (93.56)	94.33 (94.22)
RPE-BM	89.56 (89.26)	88.88 (90.71)	89.93 (89.93)	79.43 (76.04)	89.02 (89.06)	88.64 (91.61)	89.19 91.95	90.07 (90.16)
Average	94.09 (94.10)	93.46 (93.93)	93.85 (93.85)	88.48 (88.40)	93.38 (93.38)	93.78 (94.62)	94.05 (94.59)	94.32 (94.35)

Table 5
Segmentation performance on private dataset by different methods (%).

	Dice/(PA)							
	U-Net (Ronneberger et al., 2015)	ReLayNet (Roy et al., 2017)	MGUNet (Li et al., 2021)	Y-Net (Farshad et al., 2022)	EMVNet (He et al., 2023)	DA-TransUNet (Sun et al., 2024)	TransUNet (Chen et al., 2024)	Ours
Retina	95.46 (96.13)	95.64 (96.29)	95.46 (95.10)	95.11 (93.85)	92.70 (92.59)	95.71 (95.53)	94.84 (93.95)	95.37 (97.27)
Choroid	90.73 (88.33)	89.37 (86.54)	90.10 (88.79)	87.40 (84.34)	81.67 (84.30)	89.04 (86.53)	88.63 (86.91)	90.78 (89.69)
Disc	80.94 (78.91)	81.42 (76.20)	81.39 (80.36)	82.11 (78.71)	77.26 (83.33)	79.73 (76.69)	80.17 (80.82)	82.50 (78.29)
Average	89.05 (87.79)	88.81 (86.34)	88.98 (88.09)	88.21 (85.63)	83.88 (86.74)	88.16 (86.25)	87.88 (87.22)	89.55 (88.42)

respectively. It is notably that the Dice achieves optimal segmentation on all layers, while the performance of PA is satisfactory in most retinal layers.

For OCTA500 dataset, the segmentation results are shown in Table 4. It can be seen that our method achieves the best performance on Dice and ranks third on PA. For private dataset, the segmentation results are shown in Table 5. It can be seen that our method achieves the average Dice score of 89.55% and PA of 88.42%, which are 0.50% and 0.33% higher than the suboptimal methods, respectively. To summarize, our method shows the best overall performance in the above five datasets, while other comparison methods indicate inconsistent results for different segmentation tasks, which demonstrates the flexibility and robustness of our method in different scenarios.

3.3.2. Qualitative analysis

To qualitatively evaluate our method, we present the visualization results as shown in Figs. 5–8. We evaluate the segmentation results from the perspective of overall segmentation effect, continuity, and smoothness. It is obvious that our method consistently achieves the best visualization results on the five datasets. For MGU dataset, our method achieves satisfactory results both in optic disc and retinal layers. As shown in the areas circled in white, the challenging GCL layer is well segmented due to the usage of boundary and distribution prior information. Besides, some layers appear discontinuity as the red circles indicate, especially in the results of U-Net. However, our method obtains better performance in terms of the continuity and smoothness of retinal layers. For private dataset, our method accurately predicts each layer structure, especially the optic disc. In Fig. 9, the performance of our method in optic disc segmentation is better than other methods.

This shows that our method has an advantage over other methods in optic disc prediction. For NR206 dataset, RNFL, GCL+IPL and INL are extremely slender at the center of macula, which is a great challenge for models. However, our method can accurately partition the slender structures, which is mainly attributed to the reinforcement constraint involved in the distribution comparison of the proposed approach. Fig. 8 presents the segmentation results of various methods applied to fundus OCT images with severe lesions. It is evident that our method demonstrates the best performance. Although our approach does not achieve the highest PA index on OCTA500 dataset, it excels in segmenting images with severe lesions, highlighting the superiority of our method. Fig. 10 shows the segmentation results of RNFL and GCL+IPL layers at the center of macula on NR206 dataset. Green represents the correctly segmented area, red is the false negative area, and blue is the false positive area. It can be noted that EMVNet can achieve good segmentation performance in RNFL and GCL+IPL layers at the center of macula due to the addition of a large number attention modules to perceive the elongated regions. Compared with it, our method segments the retinal layers more continuously and the overall performance is the best, which verifies the excellent segmentation capability of our method on slender retinal boundaries.

3.4. Ablation studies

Due to the space limitation, we select the MGU dataset with optic disc and the DUKE dataset without optic disc for ablation study. Focal loss, position channel attention (Attention), and FEDM with reinforcement constraint (RC-FEDM), which consists of FEDM and distribution comparison module, are validated through the comparison

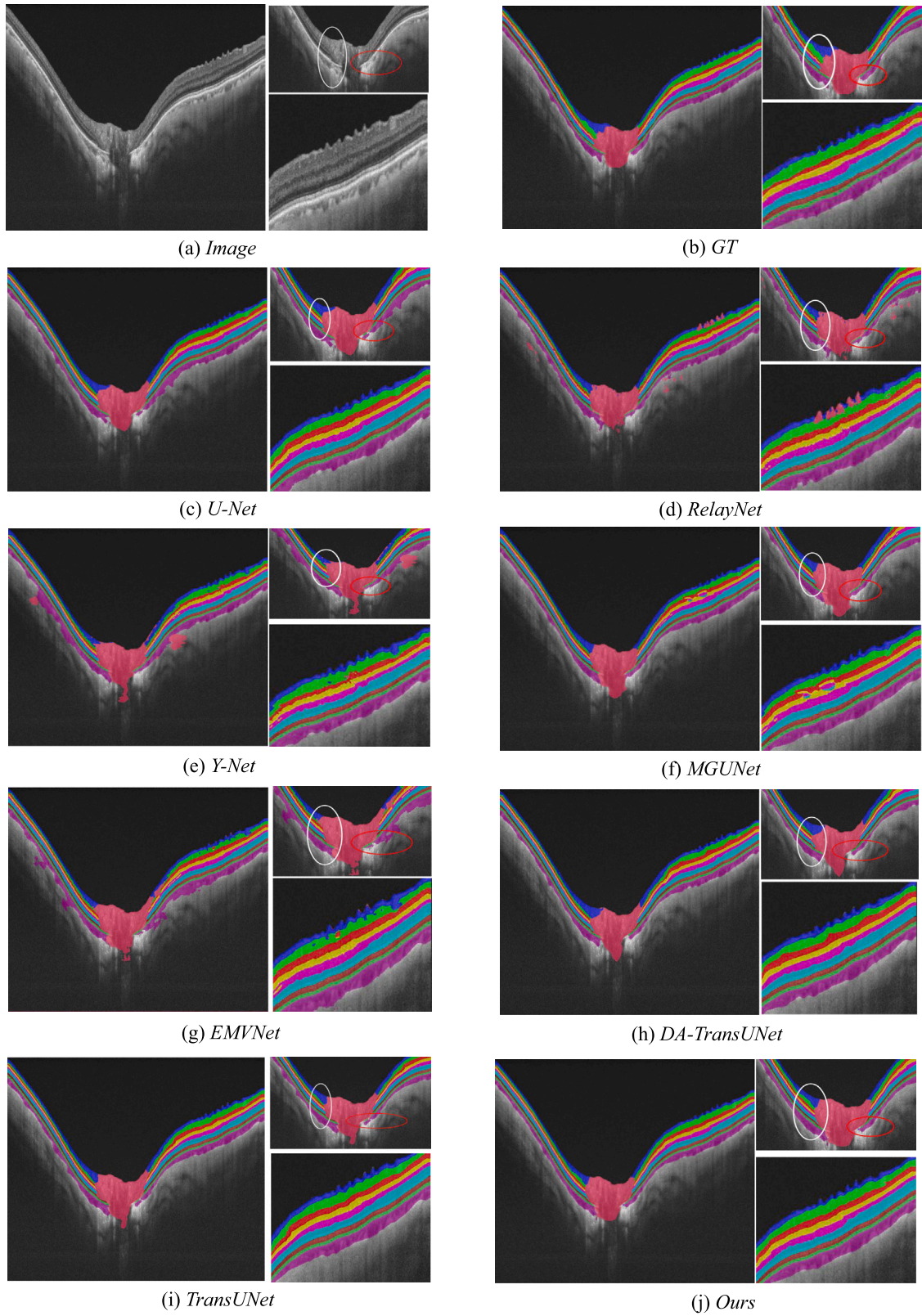


Fig. 5. Visualization of segmentation results on MGU datasets. (a) The sample image. (b) Ground truth. (c)–(i) The segmentation results of the comparison models. (j) Ours.

study. Focal loss means whether focal loss is used. Specifically, the ✓ of Focal loss means that we use the overall loss with Focal loss for training, otherwise, we replace the Focal loss with cross-entropy loss

(CEloss) for fair comparison. Attention means whether to use position channel attention. RC-FEDM means whether FEDM with reinforcement constraint is used.

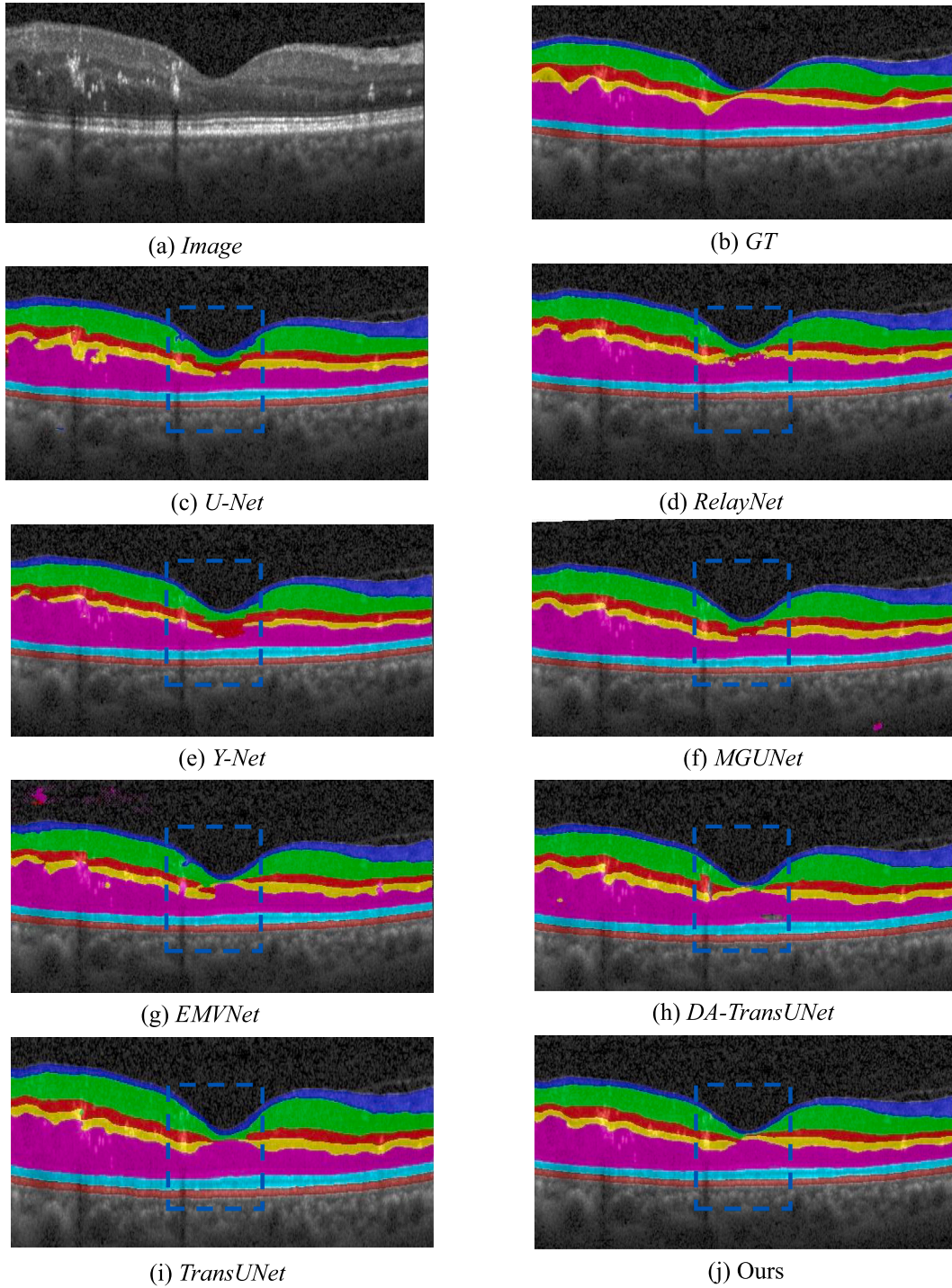


Fig. 6. Visualization of segmentation results on DUKE datasets. (a) The sample image. (b) Ground truth. (c)–(j) The segmentation results of the comparison models. (j) Ours.

3.4.1. Ablation study on MGU dataset

The ablation experimental results on MGU dataset are presented in Table 6. In general, compared with the baseline, the addition of each module contributes to the improvement in both Dice and PA. By comparing the last two columns of Table 6, it can be seen that when other conditions are the same, the effect of focal loss is better than CEloss. The segmentation performance with focal loss can be improved for most retinal layers, which verifies that the issue of imbalanced layer distribution is alleviated to some extent. Furthermore, it can be revealed from Table 6 that the position channel attention also contributes to the performance improvement of our model. This

indicates that the global spatial and channel perception plays an active role in segmentation of retinal layers. In addition, the segmentation performance is significantly promoted by adding the RC-FEDM compared with other modules, especially for the challenging GCL layer, which demonstrates that the RC-FEDM correctly handles the ambiguous boundaries of retina by reinforcing the constraint of boundary and distribution prior information. Fig. 11 shows the visual segmentation results of the ablation experiment on MGU dataset. The difficulty in segmenting the sample image mainly lies in the interference of artifacts, as shown in the red box, which is enlarged at the lower right corner. As shown in Fig. 11, it is evident that both the focal loss and RC-FEDM help to reduce the interference of artifacts.

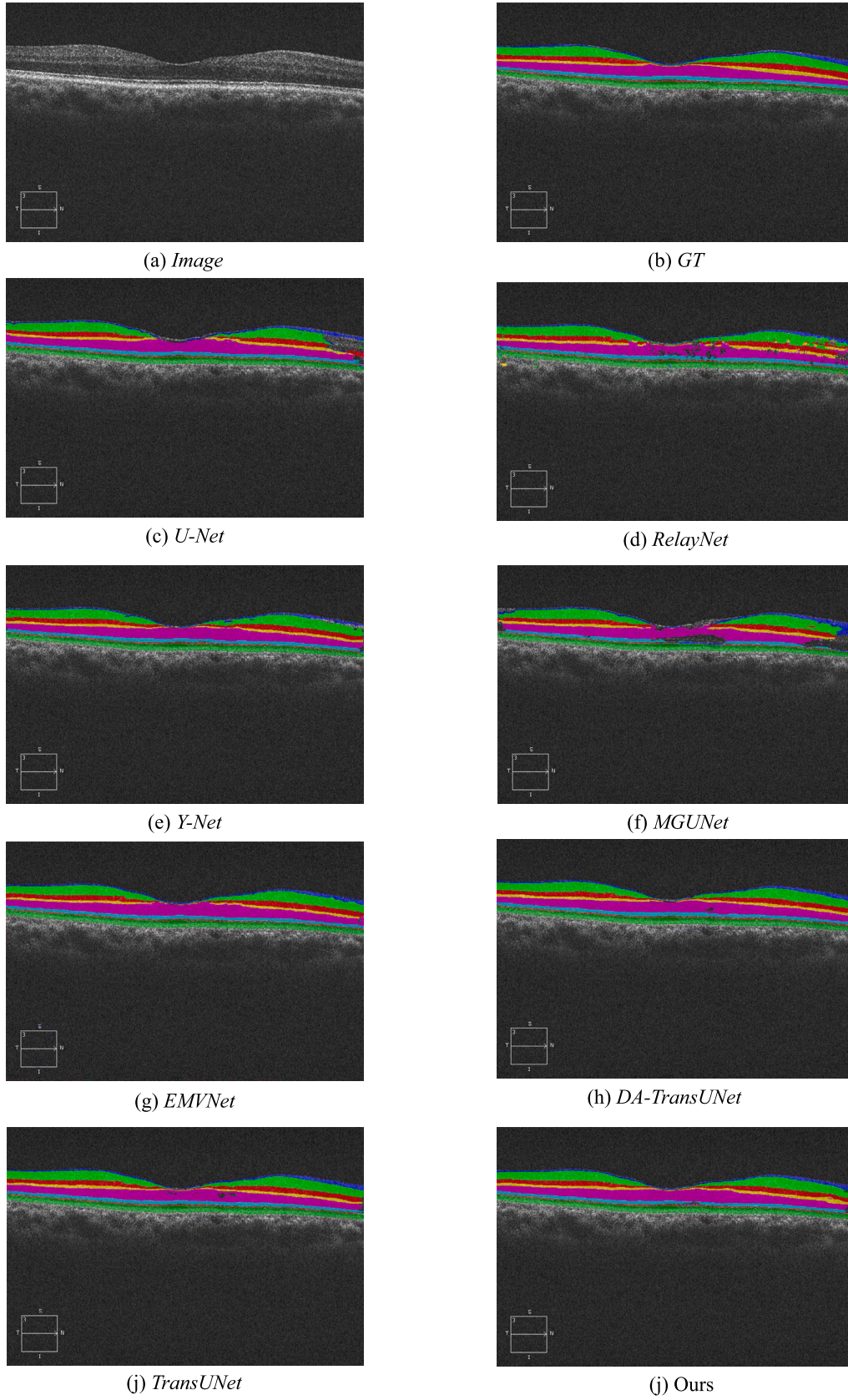


Fig. 7. Visualization of segmentation results on NR206 datasets. (a) The sample image. (b) Ground truth. (c)–(i) The segmentation results of the comparison models. (j) Ours.

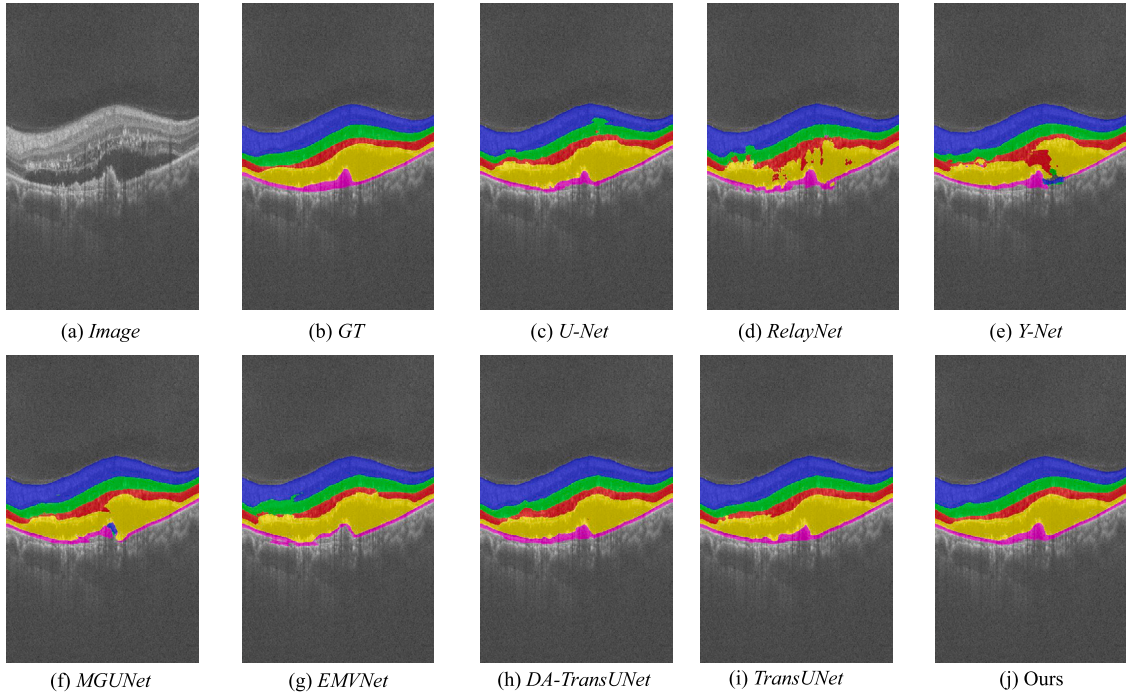


Fig. 8. Visualization of segmentation results on OCTA500 dataset. (a) The sample image. (b) Ground truth. (c)–(i) The segmentation results of the comparison models. (j) Ours.

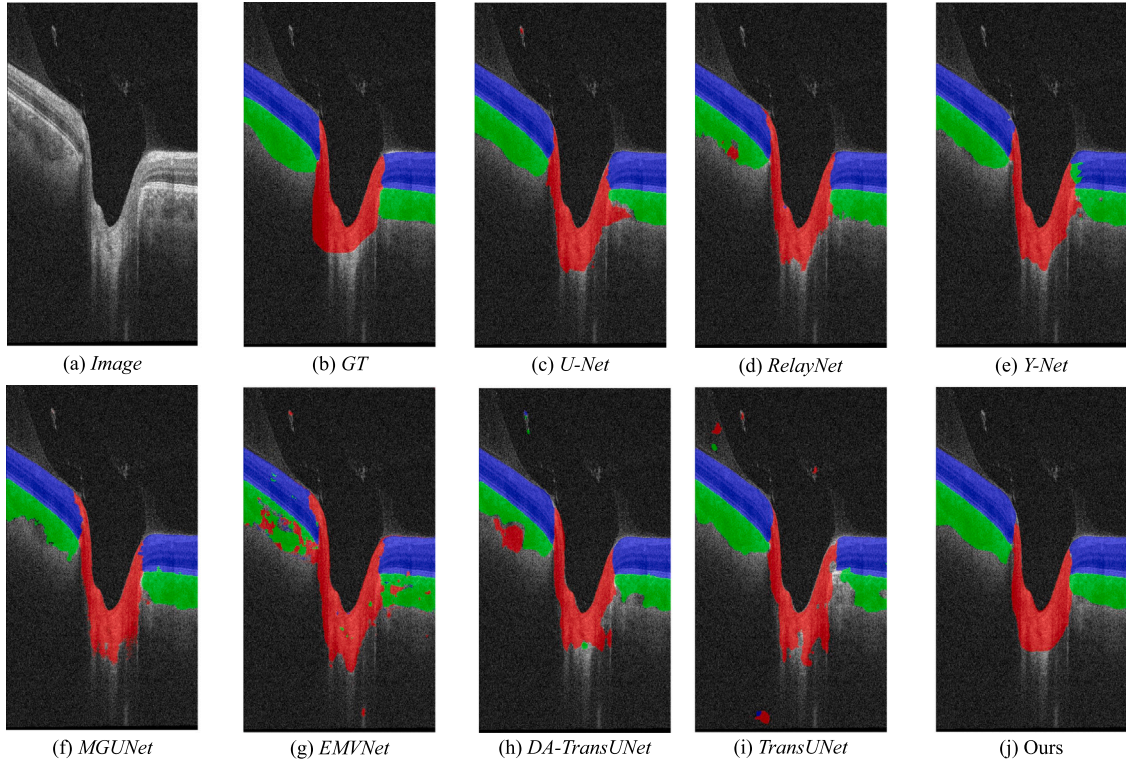


Fig. 9. Visualization of segmentation results on private dataset. (a) The sample image. (b) Ground truth. (c)–(i) The segmentation results of the comparison models. (j) Ours.

In addition, we verify the segmentation effect when different losses are used. As shown in the first column of Table 6, Dice loss and CEloss are used and regarded as the baseline. In the second column of Table 6, CEloss is replaced by focal loss. Compared with the baseline network, the overall Dice and PA increase by 0.6% and 0.57%, respectively. This further indicates that focal loss is more suitable for retinal layer segmentation tasks than CEloss. In the fourth column of Table 6, we

add the loss (L_{sim}) to the baseline loss, and the overall Dice and PA increase by 0.3% and 0.29%, respectively.

To further verify the effectiveness of incorporating label distribution in the two stages of optic disc segmentation and retinal layer segmentation, ablation experiments were conducted, as shown in Table 7. RLD refers to the retinal layer label distribution, and ODD refers to the optic disc label distribution. The results indicate that when only

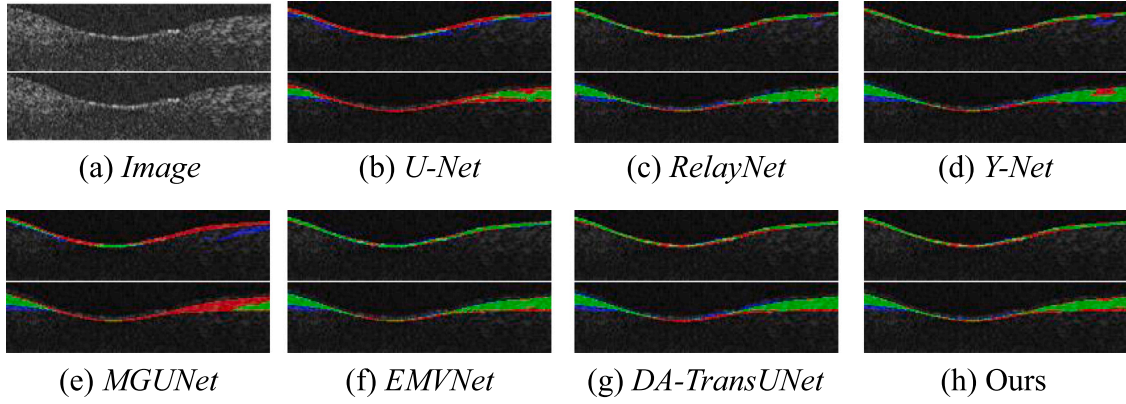


Fig. 10. The segmentation results of RNFL and GCL+IPL at the center of macula on NR206 dataset. (a) Sample images. (b)–(g) The segmentation results of the comparison models. (h) Ours. The upper part of the picture is RNFL, and the lower part is GCL+IPL.

Table 6

Ablation study on MGU dataset (%). The first row represents Dice. The second row represents PA.

Focal loss	✓				✓	✓		✓
Attention		✓			✓			✓
RC-FEDM			✓		✓	✓	✓	✓
RNFL	82.30 (82.68)	83.00 (85.06)	83.06 (83.70)	82.34 (83.33)	83.31 (83.74)	83.18 (84.82)	83.16 (84.21)	83.52 (84.59)
GCL	68.54 (68.69)	69.64 (69.50)	69.98 (69.49)	70.40 (72.25)	70.14 (69.98)	70.10 (70.22)	71.07 (71.11)	71.22 (72.34)
IPL	74.05 (72.09)	75.04 (74.49)	74.71 (75.10)	74.86 (74.09)	75.44 (75.37)	74.64 (73.46)	75.63 (75.26)	74.99 (73.53)
INL	77.88 (78.09)	78.77 (80.42)	78.05 (79.07)	78.87 (79.66)	79.41 (81.46)	78.86 (79.07)	79.66 (81.16)	79.55 (80.35)
OPL	81.28 (82.69)	81.46 (80.98)	81.23 (82.00)	81.82 (81.88)	81.84 (82.12)	81.88 (81.65)	82.20 (81.77)	82.29 (82.01)
ONL	90.68 (91.03)	91.06 (92.34)	90.86 (91.53)	90.57 (91.49)	91.09 (91.95)	91.06 (92.69)	91.18 (93.10)	91.23 (93.20)
IS/OS	86.05 (85.21)	86.46 (86.34)	86.52 (86.34)	86.32 (86.06)	86.64 (87.29)	86.59 (86.31)	86.47 (87.24)	86.57 (86.48)
RPE	82.67 (84.39)	82.76 (82.35)	82.86 (83.57)	82.52 (82.27)	82.88 (82.84)	83.08 (84.52)	82.72 (83.69)	82.95 (84.01)
Choroid	88.31 (87.02)	89.08 (89.79)	89.62 (90.82)	89.12 (88.55)	89.36 (88.52)	89.80 (90.21)	89.50 (90.03)	89.60 (90.06)
Disc	83.15 (89.28)	83.65 (85.66)	83.22 (86.29)	81.09 (84.52)	83.83 (87.03)	83.75 (87.39)	83.37 (86.67)	84.49 (88.44)
Layer	81.31 (81.32)	81.92 (82.36)	81.88 (82.40)	81.87 (82.18)	82.23 (82.59)	82.13 (82.55)	82.40 (83.06)	82.44 (82.95)
Average	81.49 (82.12)	82.09 (82.69)	82.01 (82.79)	81.79 (82.41)	82.40 (83.03)	82.29 (83.00)	82.50 (83.42)	82.64 (83.50)

Table 7

Label distribution ablation study on MGU dataset (%). The first row represents Dice. The second row represents PA.

	RNFL	GCL	IPL	INL	OPL	ONL	IS/OS	RPE	Choroid	Disc	Layer	Average
RLD	83.06 (85.08)	69.51 (69.08)	74.46 (72.74)	78.93 (80.05)	81.57 (80.46)	90.85 (91.58)	86.64 (86.60)	82.98 (83.75)	88.92 (88.21)	80.00 (85.19)	81.88 (81.95)	81.69 (82.27)
ODD	82.87 (83.09)	69.99 (70.42)	74.90 (73.79)	79.32 (81.06)	82.08 (83.19)	90.95 (91.42)	86.68 (86.82)	82.78 (82.77)	89.64 (88.92)	82.69 (86.71)	82.13 (82.48)	82.16 (82.90)
RLD+ODD	83.52 (84.59)	71.22 (72.34)	74.99 (73.53)	79.55 (80.35)	82.29 (82.01)	91.23 (93.20)	86.57 (86.48)	82.95 (84.01)	89.60 (90.06)	84.49 (88.44)	82.44 (82.95)	82.64 (83.50)

RLD is used, the model's segmentation performance is suboptimal. This is because the optic disc is adjacent to all retinal layers, and poor segmentation of the optic disc significantly impacts the segmentation of retinal layers. When only ODD is used, the segmentation of optic disc improves markedly, which leads to better overall segmentation results of the model. When both ODD and RLD are applied, the model achieves the best segmentation performance.

3.4.2. Ablation study on DUKE dataset

The ablation experimental results on DUKE dataset are detailed in Table 8. The effectiveness of focal loss, position channel attention, and RC-FEDM can also be well validated. Moreover, the RC-FEDM shows its ability to deal with retinal layers affected by the interference of lesions, as shown in Fig. 12. Due to the influence of edema, the choroid layer appears similarly to the GCL+IPL, INL, and OPL layers.

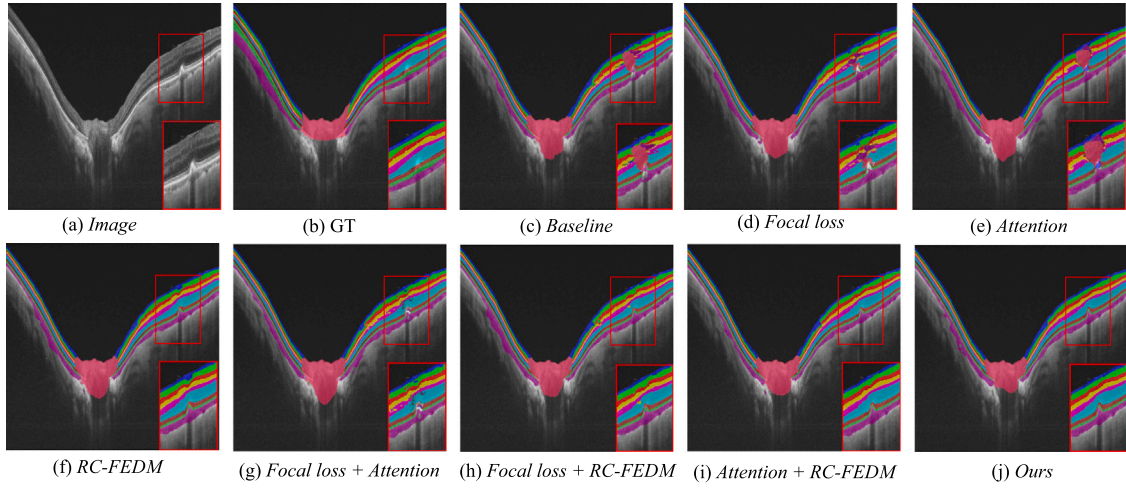


Fig. 11. The segmentation results of ablation study on MGU dataset. (a) The sample image. (b) Ground truth. (c)–(i) The segmentation results of different module combinations. (j) Ours.

Table 8

Ablation study on DUKE dataset (%). The first row represents Dice. The second row represents PA.

	✓	✓	✓	✓	✓	✓	✓	✓
Focal loss								
Attention								
RC-FEDM								
RNFL	87.09 (87.81)	87.23 (88.85)	87.14 (88.65)	87.94 (87.84)	85.95 (86.62)	88.12 (88.16)	88.18 (88.91)	87.80 (89.23)
GCL+IPL	87.07 (89.67)	89.61 (89.91)	89.30 (91.64)	90.33 (90.28)	88.91 (89.39)	90.82 (91.30)	90.40 (90.29)	91.07 (91.48)
INL	69.93 (81.18)	71.36 (82.43)	71.39 (82.55)	79.64 (81.82)	76.64 (82.66)	80.06 (81.50)	80.29 (83.46)	81.40 (84.14)
OPL	75.43 (76.41)	76.64 (79.69)	77.99 (77.66)	93.73 (80.80)	78.09 (79.14)	79.42 (81.57)	79.25 (80.63)	80.36 (81.67)
ONL+ISM	89.97 (91.71)	88.63 (91.25)	89.97 (91.31)	93.73 (93.59)	91.54 (92.27)	93.44 (92.55)	93.62 (92.33)	93.77 (92.47)
ISE	89.74 (90.52)	89.43 (90.58)	89.36 (89.94)	88.97 (89.38)	89.31 (89.77)	88.93 (88.79)	89.39 (90.92)	89.43 (89.37)
OS-RPE	88.11 (87.54)	87.74 (87.36)	88.04 (88.69)	87.65 (87.04)	87.79 (88.47)	87.76 (87.20)	87.93 (87.86)	88.13 (88.72)
Average	83.90 (86.41)	84.38 (87.11)	84.74 (87.21)	86.80 (87.25)	85.46 (86.72)	86.94 (87.30)	87.01 (87.77)	87.42 (88.15)

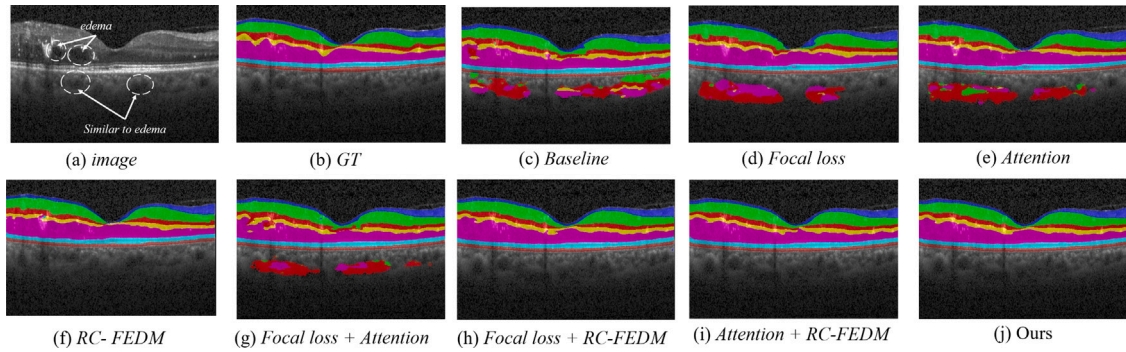


Fig. 12. The segmentation results of ablation study on DUKE dataset. (a) The sample image. (b) Ground truth. (c)–(i) The segmentation results of different module combinations. (j) Ours.

When the choroid layer suffers from a heavy black shadow caused by edema, it may be mistakenly recognized as other retinal layers, leading to segmentation errors. In addition, there are also prediction errors between adjacent layers. As shown in Fig. 12(c), the baseline network predicts part of ONL+ISM (Pink) as OPL (Yellow). After the incorporation of RC-FEDM, the network places more emphasis on the boundary of retinal layers. Through the constraint imposed on boundary and

distribution prior, the model improves the discernment between layers, leading to an overall enhancement in segmentation performance as shown in Fig. 12(f). Therefore, our method can also achieve excellent segmentation results in lesion-disturbed datasets.

Similar to the MGU dataset, we verify the segmentation effect of different losses on the DUKE dataset. As shown in Table 8, compared with the baseline network, after replacing CEloss with focal loss, the

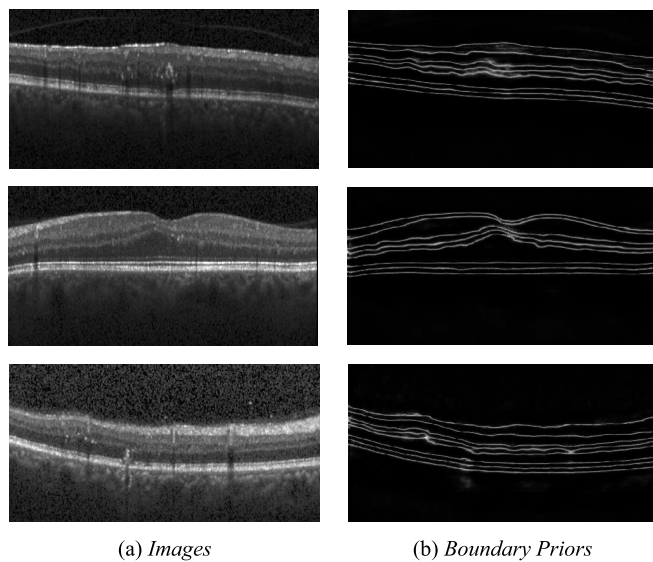


Fig. 13. Visualization of boundary prior. (a) Sample images. (b) Boundary priors.

overall Dice and PA increase by 0.48% and 0.70% respectively; after adding the loss (L_{sim}) to the baseline, the overall Dice and PA increase by 2.9% and 0.84%, respectively.

In order to verify that the boundary prior can be obtained by calculating the information entropy of the distribution prior, the obtained boundary prior is visualized, as shown in Fig. 13. It can be noted that the boundary prior indicates the retinal layer boundaries clearly, which demonstrates the rationality of obtaining boundary prior through information entropy.

4. Conclusion

In this study, motivated by the challenges that existing methods cannot achieve consistent segmentation performance on different types of datasets, such as retinal OCT images with optic disc and lesion interference, we design a general model for retinal layer segmentation. Specifically, the feature enhanced decoding module with reinforcement constraint is proposed to obtain more continuous and smoother boundaries through distribution comparison of prior information. Moreover, the position channel attention is introduced to obtain global dependencies of both space and channel. Furthermore, the focal loss is introduced to handle the imbalanced distribution of retinal OCT images. The experimental results on five datasets demonstrate that our method has the state-of-the-art overall performance. In future studies, we aim to combine the proposed method with unsupervised learning that is more adapted to clinical needs.

CRedit authorship contribution statement

Jinbao Hao: Writing – original draft, Methodology. **Huiqi Li:** Validation, Resources, Funding acquisition. **Shuai Lu:** Formal analysis, Conceptualization. **Zeheng Li:** Software, Formal analysis. **Weihang Zhang:** Writing – review & editing, Supervision, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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